

- 7 (a) Define *magnetic flux density*.

Magnetic flux density is (numerically equal to) the force per unit length on straight

 conductor carrying unit current normal to the field.

[2]

- (b) A rigid square single-wire coil PQRS of side 4.00 cm and mass 25.0 g hangs vertically in a uniform magnetic field with magnetic flux density of 0.0420 T acting vertically downwards, as shown in Fig. 7.1

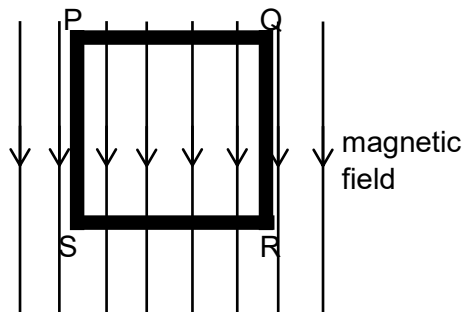


Fig. 7.1

This coil can rotate freely, pivoted along the top edge PQ. When a current of 7.50 A passes through the coil, it hangs at an angle of θ to the vertical as shown in the side view of the coil in Fig. 7.2.

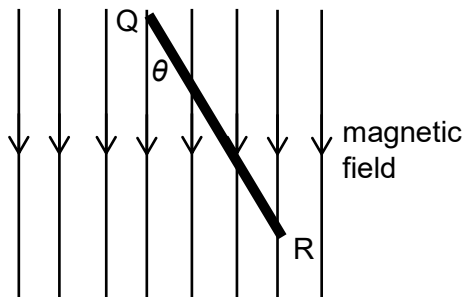


Fig. 7.2

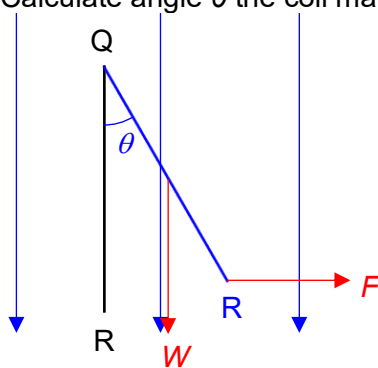
- (i) State the direction of current along the side RS.

direction = from S to R [1]

- (ii) Explain how the answer in (i) was determined.

.....
 [1]

- (iii) Calculate angle θ the coil makes with the magnetic field.



side view

By principle of moments,

$$\tau_{\text{net}} = 0$$

$$F(L \cos \theta) = W \left(\frac{L}{2} \sin \theta \right)$$

$$2BIL = mg \tan \theta$$

$$\tan \theta = \frac{2BIL}{mg}$$

$$\theta = \tan^{-1} \frac{2BIL}{mg}$$

$$= \tan^{-1} \left[\frac{2(0.0420)(7.50)(4.00 \times 10^{-2})}{(25.0 \times 10^{-3})(9.81)} \right]$$

$$= 5.87^\circ$$

$\theta = \dots\dots\dots [3]$

[Total: 7]